



# **GCE A LEVEL MARKING SCHEME**

**SUMMER 2024**

**A LEVEL  
CHEMISTRY – UNIT 3  
1410U30-1**

### **About this marking scheme**

The purpose of this marking scheme is to provide teachers, learners, and other interested parties, with an understanding of the assessment criteria used to assess this specific assessment.

This marking scheme reflects the criteria by which this assessment was marked in a live series and was finalised following detailed discussion at an examiners' conference. A team of qualified examiners were trained specifically in the application of this marking scheme. The aim of the conference was to ensure that the marking scheme was interpreted and applied in the same way by all examiners. It may not be possible, or appropriate, to capture every variation that a candidate may present in their responses within this marking scheme. However, during the training conference, examiners were guided in using their professional judgement to credit alternative valid responses as instructed by the document, and through reviewing exemplar responses.

Without the benefit of participation in the examiners' conference, teachers, learners and other users, may have different views on certain matters of detail or interpretation. Therefore, it is strongly recommended that this marking scheme is used alongside other guidance, such as published exemplar materials or Guidance for Teaching. This marking scheme is final and will not be changed, unless in the event that a clear error is identified, as it reflects the criteria used to assess candidate responses during the live series.

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## WJEC GCE A LEVEL CHEMISTRY UNIT 3

### SUMMER 2024 MARK SCHEME

#### GENERAL INSTRUCTIONS

##### Extended response questions

A level of response mark scheme is applied. The complete response should be read in order to establish the most appropriate band. Award the higher mark if there is a good match with content and communication criteria. Award the lower mark if either content or communication barely meets the criteria.

##### Marking rules

All work should be seen to have been marked.

Marking schemes will indicate when explicit working is deemed to be a necessary part of a correct answer.

Crossed out responses not replaced should be marked.

##### Marking abbreviations

The following may be used in marking schemes or in the marking of scripts to indicate reasons for the marks awarded.

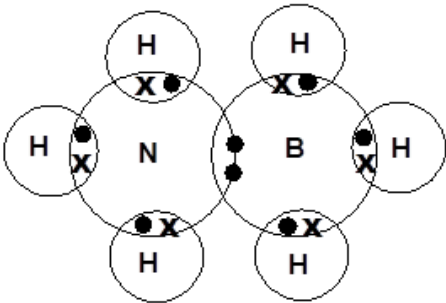
|     |   |                       |
|-----|---|-----------------------|
| cao | = | correct answer only   |
| ecf | = | error carried forward |
| bod | = | benefit of doubt      |

Credit will be given for correct and relevant alternative responses which are not recorded in the mark scheme.

## SECTION A

| Question |     |  |  | Marking details                                                                                                                                                                                            | Marks available |          |          |           |          |          |
|----------|-----|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|----------|
|          |     |  |  |                                                                                                                                                                                                            | AO1             | AO2      | AO3      | Total     | Maths    | Prac     |
| 1        | (a) |  |  | $\text{Mg}^{2+}(\text{g}) + 2\text{Cl}^{-}(\text{g}) \rightarrow \text{MgCl}_2(\text{s})$                                                                                                                  |                 | 1        |          | 1         |          |          |
|          | (b) |  |  | standard enthalpy change of hydration                                                                                                                                                                      | 1               |          |          | 1         |          |          |
| 2        | (a) |  |  | oxidation state of chlorine at start is 0; at end it is $-1$ (in NaCl) and $+1$ (in NaOCl) (1)<br><br>chlorine (or same element) has been both oxidised and reduced (1)                                    | 1               | 1        |          | 2         |          |          |
|          | (b) |  |  | bleach                                                                                                                                                                                                     | 1               |          |          | 1         |          |          |
| 3        |     |  |  | $[\text{H}^{+}] = \frac{1.00 \times 10^{-14}}{2 \times 2.34 \times 10^{-2}} = 2.13 \times 10^{-13} \text{ mol dm}^{-3}$ (1)<br><br>$\text{pH} = -\log 2.13 \times 10^{-13} = 12.7$ (1)<br><br>ecf possible |                 | 2        |          | 2         | 2        |          |
| 4        |     |  |  | more efficient release of energy from fuel                                                                                                                                                                 | 1               |          |          | 1         |          |          |
| 5        |     |  |  | $+3$ (must give reason to gain this mark) (1)<br><br>inert pair effect increases down the group (so oxidation state of [group number $- 2$ ] is more stable) (1)                                           | 1               | 1        |          | 2         |          |          |
|          |     |  |  | <b>Section A total</b>                                                                                                                                                                                     | <b>5</b>        | <b>5</b> | <b>0</b> | <b>10</b> | <b>2</b> | <b>0</b> |

## SECTION B

| Question |     |      |    | Marking details                                                                                                                                                                                        | Marks available |     |     |       |       |      |
|----------|-----|------|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |    |                                                                                                                                                                                                        | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 6        | (a) | (i)  |    | base is a proton acceptor (1)<br><br>lone pair on nitrogen (can form coordinate bond with H <sup>+</sup> ) (1)                                                                                         | 2               |     |     | 2     |       |      |
|          |     | (ii) | I  | award (1) for any of following<br>ammonium salt of any strong acid e.g. ammonium sulfate,<br>ammonium nitrate, ammonium chloride<br>any strong acid e.g. sulfuric acid, nitric acid, hydrochloric acid | 1               |     |     | 1     |       |      |
|          |     |      | II | award (1) for any sensible answer e.g.<br>using enzymes<br>storage of biological molecules<br>fermentation                                                                                             | 1               |     |     | 1     |       | 1    |
|          | (b) | (i)  |    |  <p>accept correct versions using triangles (or other symbols) to differentiate between electrons</p>                | 1               |     |     | 1     |       |      |

| Question  |                                                                               |                                                                                                                                                       |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Marks available |                                                         |                                                         |           |                                            |                                                                                                                                                       |         |                                                                               |                                                                                                                         |   |  |  |   |  |  |
|-----------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|---------------------------------------------------------|---------------------------------------------------------|-----------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|---|--|--|---|--|--|
|           |                                                                               |                                                                                                                                                       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | AO1             | AO2                                                     | AO3                                                     | Total     | Maths                                      | Prac                                                                                                                                                  |         |                                                                               |                                                                                                                         |   |  |  |   |  |  |
|           |                                                                               | (ii)                                                                                                                                                  |  | award (2) for <b>all</b> three boxes correct<br>award (1) for any <b>two</b> boxes correct<br><table><tr><td></td><td>Similarity between hexagonal boron nitride and graphite</td><td>Difference between hexagonal boron nitride and graphite</td></tr><tr><td>Structure</td><td>layers of hexagonal sheets present in both</td><td>layers in register in boron nitride and out of register in graphite / atoms of adjacent layers above one another in boron nitride but not in graphite</td></tr><tr><td>Bonding</td><td>each atom is bonded by covalent bonds to three others in both graphite and BN</td><td>delocalised electrons in graphite but not in boron nitride / bonds are non-polar in graphite but polar in boron nitride</td></tr></table> |                 | Similarity between hexagonal boron nitride and graphite | Difference between hexagonal boron nitride and graphite | Structure | layers of hexagonal sheets present in both | layers in register in boron nitride and out of register in graphite / atoms of adjacent layers above one another in boron nitride but not in graphite | Bonding | each atom is bonded by covalent bonds to three others in both graphite and BN | delocalised electrons in graphite but not in boron nitride / bonds are non-polar in graphite but polar in boron nitride | 2 |  |  | 2 |  |  |
|           | Similarity between hexagonal boron nitride and graphite                       | Difference between hexagonal boron nitride and graphite                                                                                               |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                 |                                                         |                                                         |           |                                            |                                                                                                                                                       |         |                                                                               |                                                                                                                         |   |  |  |   |  |  |
| Structure | layers of hexagonal sheets present in both                                    | layers in register in boron nitride and out of register in graphite / atoms of adjacent layers above one another in boron nitride but not in graphite |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                 |                                                         |                                                         |           |                                            |                                                                                                                                                       |         |                                                                               |                                                                                                                         |   |  |  |   |  |  |
| Bonding   | each atom is bonded by covalent bonds to three others in both graphite and BN | delocalised electrons in graphite but not in boron nitride / bonds are non-polar in graphite but polar in boron nitride                               |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                 |                                                         |                                                         |           |                                            |                                                                                                                                                       |         |                                                                               |                                                                                                                         |   |  |  |   |  |  |
|           | (c)                                                                           |                                                                                                                                                       |  | phosphorus has available / accessible d-orbitals (1)<br>whilst nitrogen does not (1)<br><br>or<br><br>phosphorus can expand its octet (1)<br>but nitrogen cannot (1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2               |                                                         |                                                         | 2         |                                            |                                                                                                                                                       |         |                                                                               |                                                                                                                         |   |  |  |   |  |  |
|           |                                                                               |                                                                                                                                                       |  | Question 6 total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 9               | 0                                                       | 0                                                       | 9         | 0                                          | 0                                                                                                                                                     |         |                                                                               |                                                                                                                         |   |  |  |   |  |  |

| Question |     |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                        | Marks available |     |     |       |       |      |
|----------|-----|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                        | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 7        | (a) |  |  | <p>5 cm<sup>3</sup> must give reason</p> <p>to keep total volume constant / to ensure concentration of hydrogensulfate(IV) stays constant</p>                                                                                                                                                                                                                                                                                                          |                 |     | 1   | 1     |       | 1    |
|          | (b) |  |  | <p>first order (1) must attempt to show reasoning</p> <p>award (1) for suitable explanation/reasoning e.g.</p> <ul style="list-style-type: none"> <li>when concentration (of iodate) is doubled the time taken halves</li> <li>time taken is inversely proportional to concentration (of iodate)</li> <li>rate goes from 0.006 to 0.012 when concentration (of iodate) doubles</li> <li>rate doubles when concentration (of iodate) doubles</li> </ul> |                 | 1   | 1   | 2     | 2     | 2    |
|          | (c) |  |  | <p>award (1) for suitable rate equation with overall order of four e.g.</p> <p>rate = <math>k [\text{IO}_3^-] [\text{HSO}_3^-]^3</math></p>                                                                                                                                                                                                                                                                                                            |                 |     | 2   | 2     |       |      |
|          | (d) |  |  | <p>award (1) for any balanced equation with <math>\text{IO}_3^-</math> / <math>\text{HSO}_3^-</math> as reactants e.g.</p> <p><math>\text{IO}_3^- + 3\text{HSO}_3^- \rightarrow \text{I}^- + 3\text{HSO}_4^-</math></p> <p>ecf possible from part (c)</p>                                                                                                                                                                                              |                 |     | 1   | 1     |       |      |

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                      | Marks available |     |     |       |       |      |
|----------|-----|------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |  |                                                                                                                                                                                                                                                                                      | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          | (e) | (i)  |  | bright/canary yellow                                                                                                                                                                                                                                                                 | 1               |     |     | 1     |       | 1    |
|          |     | (ii) |  | moles of gas at 200°C = $5.653 \times 10^{-3}$ mol (1)<br>moles of gas at 20°C = $4.562 \times 10^{-3}$ mol (1)<br>[if the same error made in both calculations above then penalise once only]<br>moles of iodine = $1.091 \times 10^{-3}$ mol<br>⇒ percentage of iodine = 19.3% (1) |                 |     | 3   | 3     | 3     |      |
|          |     |      |  | Question 7 total                                                                                                                                                                                                                                                                     | 1               | 1   | 8   | 10    | 5     | 4    |



| Question |     |      |    | Marking details                                                                                                                                                                                                                                                        | Marks available |     |     |       |       |      |
|----------|-----|------|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |    |                                                                                                                                                                                                                                                                        | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 8        | (a) |      |    | award (1) for either of following<br>energy of the <i>d</i> -electrons are similar<br>successive ionisation energies are similar                                                                                                                                       | 1               |     |     | 1     |       |      |
|          | (b) | (i)  |    | blue (1)<br><br>the frequency of light that is <u>not absorbed</u> is in this region (1)                                                                                                                                                                               |                 | 2   |     | 2     |       |      |
|          |     | (ii) | I  | a (small) molecule/ion with a lone pair (that can bond to a transition metal ion)                                                                                                                                                                                      | 1               |     |     | 1     |       |      |
|          |     |      | II | award (1) for any of following<br>lone pair from ligand combines with empty orbital<br>lone pair from ligand can combine with electron deficient transition metal atom<br>sharing of a pair of electrons which are both from ligand<br><br>forms a coordinate bond (1) | 2               |     |     | 2     |       |      |
|          | (c) | (i)  |    | $2\text{I}^- \rightleftharpoons \text{I}_2 + 2\text{e}^-$                                                                                                                                                                                                              |                 | 1   |     | 1     |       |      |
|          |     | (ii) |    | $\text{VO}_2^+ + 2\text{H}^+ + \text{e}^- \rightleftharpoons \text{VO}^{2+} + \text{H}_2\text{O}$                                                                                                                                                                      |                 | 1   |     | 1     |       |      |

| Question |     |      |     | Marking details                                                                                                                                                                                                                                                                                                                                 | Marks available |           |          |           |          |          |
|----------|-----|------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------|----------|-----------|----------|----------|
|          |     |      |     |                                                                                                                                                                                                                                                                                                                                                 | AO1             | AO2       | AO3      | Total     | Maths    | Prac     |
|          | (d) | (i)  |     | catalyst in a different physical state from the reactants / reaction                                                                                                                                                                                                                                                                            | 1               |           |          | 1         |          |          |
|          |     | (ii) | I   | $[2 \times \Delta_f H(\text{SO}_3)] - [2 \times \Delta_f H(\text{SO}_2)] = -197$ (1)<br>$\Delta_f H(\text{SO}_3) = -395.5 \text{ kJ mol}^{-1}$ (1)                                                                                                                                                                                              |                 | 2         |          | 2         | 2        |          |
|          |     |      | II  | $T = \frac{\Delta H}{\Delta S}$ (1)<br>$T = 1053 \text{ K}$ (1)<br>student is incorrect must give reason<br>$\Delta G$ is negative below this temperature so the reaction is feasible at a lower temperature (1)                                                                                                                                |                 | 2         |          |           |          |          |
|          |     |      |     | $\Delta G$ is negative below this temperature so the reaction is feasible at a lower temperature (1)                                                                                                                                                                                                                                            |                 |           | 1        | 3         | 2        |          |
|          |     |      | III | initial concentrations of $\text{SO}_2$ and $\text{O}_2 = 0.020 \text{ mol dm}^{-3}$ (1)<br>equilibrium concentrations – both needed for (1)<br>$[\text{SO}_2] = 0.006 \text{ mol dm}^{-3}$<br>$[\text{O}_2] = 0.013 \text{ mol dm}^{-3}$<br>$K_c = \frac{(0.014)^2}{(0.006)^2 \times (0.013)} = 419$ (1)<br>$\text{mol}^{-1} \text{ dm}^3$ (1) |                 | 4         |          | 4         | 4        |          |
|          |     |      |     | <b>Question 8 total</b>                                                                                                                                                                                                                                                                                                                         | <b>5</b>        | <b>12</b> | <b>1</b> | <b>18</b> | <b>8</b> | <b>0</b> |

| Question |     |      |    | Marking details                                                                                                                                                                                                                                                            | Marks available |     |     |       |       |      |
|----------|-----|------|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |    |                                                                                                                                                                                                                                                                            | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 9        | (a) | (i)  |    | all <b>three</b> needed<br>chloride – white<br>bromide – cream (accept off-white)<br>yellow – iodide                                                                                                                                                                       | 1               |     |     | 1     |       | 1    |
|          |     | (ii) |    | $\Delta_f H^\ominus (\text{AgBr}) = 105.6 + (\frac{1}{2} \times -243.1) - 84.4 \quad (1)$<br>$\Delta_f H^\ominus (\text{AgBr}) = -100.35 \quad (1)$<br>AgCl more stable than AgBr (ecf possible from values) (1)<br>as standard enthalpy of formation is more negative (1) |                 | 2   | 2   | 4     | 1     |      |
|          | (b) | (i)  | I  | $\text{Cu} + 2\text{Ag}^+ \rightarrow \text{Cu}^{2+} + 2\text{Ag}$                                                                                                                                                                                                         |                 | 1   |     | 1     |       |      |
|          |     |      | II | $E^\ominus (\text{Cu})$ must be more negative/less positive than +0.80V (1)<br>award (1) for either of following reasons<br>reaction is feasible when EMF is positive<br>reaction is feasible when reduction half-cell is more positive than<br>oxidation half-cell        |                 |     | 2   | 2     |       |      |

| Question |  |      |    | Marking details                                                                                                                                                                                                                                                                                       | Marks available |     |     |       |       |      |
|----------|--|------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |  |      |    |                                                                                                                                                                                                                                                                                                       | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          |  | (ii) | I  | award (2) for correct answer<br>$E^\ominus = 0.80 - 0.93 = -0.13 \text{ V}$<br><br>if incorrect award (1) for use of 0.93 V                                                                                                                                                                           |                 | 1   | 1   | 2     | 1     |      |
|          |  |      | II | as concentration of $\text{Ag}^+$ decreases the $\text{Ag}^+ + \text{e}^- \rightleftharpoons \text{Ag}$ equilibrium shifts to the left (1)<br><br>this makes $E$ for this reaction less positive (1)<br><br>EMF of the cell is $E_{\text{Ag}} - E_{\text{Pb}}$ so the value becomes less positive (1) |                 |     | 3   | 3     |       |      |
|          |  |      |    | Question 9 total                                                                                                                                                                                                                                                                                      | 1               | 4   | 8   | 13    | 2     | 1    |

| Question |     |       |     | Marking details                                                                                                                                                                                                                              | Marks available |     |     |       |       |      |
|----------|-----|-------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |     |                                                                                                                                                                                                                                              | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 10       | (a) |       |     | $[H^+] = 10^{-3.24} \text{ (1)}$<br>$[H^+] = 5.75 \times 10^{-4} \text{ mol dm}^{-3} \text{ (1)}$                                                                                                                                            |                 | 2   |     | 2     | 2     |      |
|          | (b) | (i)   |     | filtration                                                                                                                                                                                                                                   |                 | 1   |     | 1     |       | 1    |
|          |     | (ii)  | I   | excess sodium hydroxide dissolves any precipitate produced by zinc (1)<br>excess ensures that <b>all</b> iron ions are precipitated (1)                                                                                                      |                 |     | 2   | 2     |       | 2    |
|          |     |       | II  | $\text{moles Fe}_2\text{O}_3 = \frac{14.5 \times 10^{-3}}{159.6} = 9.085 \times 10^{-5} \text{ mol} \quad (1)$<br>$\text{concentration} = \frac{2 \times (9.085 \times 10^{-5})}{0.025} = 7.27 \times 10^{-3} \text{ mol dm}^{-3} \quad (1)$ |                 | 2   |     | 2     | 2     |      |
|          |     |       | III | $\frac{0.2}{14.5} \times 100 = 1.4\%$                                                                                                                                                                                                        |                 | 1   |     | 1     | 1     |      |
|          |     | (iii) | I   | solution turns (pale) pink<br>do not accept purple to pink                                                                                                                                                                                   | 1               |     |     | 1     |       | 1    |
|          |     |       | II  | $\text{moles of MnO}_4^- = 3.80 \times 10^{-5} \text{ mol} \quad (1)$<br>$[\text{Fe}^{2+}] = 7.60 \times 10^{-3} \text{ mol dm}^{-3} \quad (1)$<br>ecf possible                                                                              |                 | 2   |     | 2     | 2     |      |

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Marks available |     |     |       |       |      |
|----------|-----|------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          |     | (iv) |  | award (1) for percentage difference between the two values<br>$\Rightarrow$ either 4.3% or 4.5% (1)<br><br>students' decision is <b>not</b> valid<br>award (1) for sensible explanation e.g. <ul style="list-style-type: none"> <li>total of percentage errors for both methods is 2.2% (1.4% + 0.8%) and the percentage difference is greater than this</li> <li>total of percentage errors in titration and gravimetric analysis allow a difference of only <math>0.16 \times 10^{-3}</math></li> <li>values are <math>7.60 \pm 0.10 \times 10^{-3}</math> and <math>7.27 \pm 0.06 \times 10^{-3}</math> and these ranges do not overlap</li> </ul>                                                                                                                                                                                                                                                   |                 |     | 2   | 2     | 2     |      |
|          | (c) | (i)  |  | royal blue / dark blue<br><br>do not accept blue / pale blue or any reference to a precipitate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1               |     |     | 1     |       | 1    |
|          |     | (ii) |  | <b>Indicative content</b> <ol style="list-style-type: none"> <li>A negative <math>\Delta G</math> means products are more likely to be present than reactants</li> <li>Higher product concentration leads to large value of <math>K_c</math></li> <li>Formation of the Cu-cyclen complex breaks four Cu-ammonia bonds and forms four Cu-cyclen bonds</li> <li>As the bonds broken and bonds made are of similar strengths enthalpy change will be small</li> <li>For a negative <math>\Delta G</math>, <math>\Delta S</math> must be positive OR <math>T\Delta S</math> must be greater than <math>\Delta H</math> as <math>\Delta G = \Delta H - T\Delta S</math></li> <li>The forward reaction produces more (aqueous) molecules as four <math>\text{NH}_3</math> are released</li> <li>The entropy of free ammonia would be higher than cyclen so <math>\Delta S</math> would be positive</li> </ol> |                 | 2   | 4   | 6     | 1     |      |

| Question |  |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Marks available |     |     |       |       |      |
|----------|--|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |  |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          |  |  |  | <p><b>5-6 marks</b><br/>The candidate includes five points with valid discussion of entropy, enthalpy and the position of equilibrium<br/><i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The information included in the response is relevant to the argument.</i></p> <p><b>3-4 marks</b><br/>The candidate includes four points with valid discussion of two of entropy, enthalpy and the position of equilibrium<br/><i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. Mainly relevant information is included in the response but there may be some minor errors or the inclusion of some information not relevant to the argument.</i></p> <p><b>1-2 marks</b><br/>The candidate includes two relevant points<br/><i>There is a basic line of reasoning which is not coherent, supported by limited evidence and with very little structure. There may be significant errors or the inclusion of information not relevant to the argument.</i></p> <p><b>0 marks</b><br/>No attempt made or no response worthy of credit.</p> |                 |     |     |       |       |      |
|          |  |  |  | Question 10 total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2               | 10  | 8   | 20    | 10    | 5    |

### UNIT 3

#### SUMMARY OF MARKS ALLOCATED TO ASSESSMENT OBJECTIVES

| Question  | AO1 | AO2 | AO3 | TOTAL MARK | MATHS | PRAC |
|-----------|-----|-----|-----|------------|-------|------|
| Section A | 5   | 5   | 0   | 10         | 2     | 0    |
| 6         | 9   | 0   | 0   | 9          | 0     | 0    |
| 7         | 1   | 1   | 8   | 10         | 5     | 4    |
| 8         | 5   | 12  | 1   | 18         | 8     | 0    |
| 9         | 1   | 4   | 8   | 13         | 2     | 1    |
| 10        | 2   | 10  | 8   | 20         | 10    | 5    |
| Total     | 23  | 32  | 25  | 80         | 27    | 10   |